Tharun Sridhar

Backend Engineer | SDE | Bengaluru, India
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Summary

Backend engineer (B.Tech CSE, 2026) building and deploying REST APIs and backend services in Django and FastAPI: JWT authentication and role-based access control, PostgreSQL data modeling with row-level locking for concurrency-safe workflows, Redis caching, and Celery background processing. Comfortable owning a service end to end, from schema design through Dockerized deployment.

Technical Skills

Languages: Python, SQL, Java
CS Fundamentals: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks
Backend: Django, Django REST Framework, FastAPI, REST API Design, JWT Authentication & RBAC, SQLAlchemy, Alembic, Celery, django-filter, drf-spectacular
Databases: PostgreSQL, SQLite, Redis
DevOps & Cloud: Docker, GitHub Actions (CI/CD), AWS (EC2, ALB, VPC, IAM), Railway, Git, Linux
Testing: pytest, pytest-django
Frontend: HTML, CSS

Education

Vellore Institute of Technology — Vellore, India
Bachelor of Technology in Computer Science — *May 2026*

Projects

Inventra *(Team Project – Lead)* | *Django, DRF, PostgreSQL, Redis, Celery, JWT, Docker* — GitHub | Live Demo

- Led a team building a role-based inventory system (Admin/Manager/Employee JWT RBAC, rate limited per scope); personally built the JWT auth (access + DB-blacklisted refresh tokens) and the REST API layer across 13 routers.
- Enforced an append-only stock ledger two ways (no write route registered, plus Django Admin edits hard-disabled) and made purchase/sale completion idempotent: a retried request returns `409` with zero side effects instead of double-applying a stock change.
- Closed a concurrency bug with PostgreSQL row-level locking (`select_for_update`): Locust testing at 40 concurrent users showed 16 silently lost stock updates with zero HTTP errors when unlocked, 0 when locked.
- Redis-cached reports with version-counter invalidation (cache hits collapse 9 queries to 1, verified via query-count assertions) and Celery background jobs for async invoices, low-stock alerts, and nightly reconciliation; deployed on AWS behind an ALB with GitHub Actions CI running pytest and ruff.

PhotoShare API | *Django, DRF, PostgreSQL, Redis, Celery, JWT* — GitHub

- Rewrote the project's original async FastAPI/SQLAlchemy backend as a sync Django + DRF port with an identical API contract and equivalent 20-test suite, trading async concurrency ceiling for one ORM everywhere and zero async-runtime bug classes.
- Hand-rolled a JWT layer on PyJWT instead of SimpleJWT's access/refresh design to match the app's single stateless-bearer-token contract; verify/reset tokens embed a password-hash fingerprint so a password change instantly invalidates any outstanding token.
- Redis cache-aside on feed/post-detail reads, Celery background thumbnail generation sharing the same Django ORM, and ownership checked against the row being deleted (403 for non-owners); Dockerized and deployed to Railway with GitHub Actions running pytest and ruff on every push.

Finvoro | *Django, DRF, PostgreSQL, JWT, pytest* — GitHub

- Built a personal finance management API (accounts, categories, transactions, budgets, reports) on Django + DRF and PostgreSQL, with every endpoint JWT-authenticated and scoped to the authenticated user.
- Implemented filterable, paginated transaction endpoints (category, account, type, date range) and computed account balances and budget totals directly from transaction data rather than storing duplicated aggregates.
- Documented the full versioned API (`/api/v1/`) with drf-spectacular (Swagger/ReDoc) and covered it with a 48-test pytest/pytest-django suite (0 failures, 96% coverage of app code); deployed via gunicorn and whitenoise, no Docker or paid infra needed.

Open Source Contributions

django-guardian/django-guardian – fixed AUTO_PREFETCH permission bug for direct-permission models #1030
drkrillo/good-first-issues – CI smoke test for the action itself #172
Ayorinha/ayorai-vision-intelligence – restored corrupted CI workflow YAML #26
Ayorinha/ayorai-vision-intelligence – MCP least-privilege regression tests #23

Certifications

AWS Solutions Architect Associate (SAA-C03) Specialization — Packt

Achievements

5⋆ HackerRank: Problem Solving, Java, Python, SQL — hackerrank
250+ LeetCode problems solved — leetcode